PROJECT REPORT

IoT-Based Smart Electrical Muscle Stimulation (EMS) and Real-Time Patient Monitoring System for Paralysis Rehabilitation

Submitted by

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Project type: Educational prototype for sensor monitoring, embedded control, and IoT data display

Note: This report describes a prototype concept. It is not a clinically validated medical device and must not be used to deliver electrical stimulation to a person.

ABSTRACT

Paralysis and prolonged immobility can reduce voluntary muscle activity and may contribute to muscle weakness and atrophy. Rehabilitation can involve supervised exercises and, in selected clinical situations, electrical muscle stimulation. This project explores an Internet of Things (IoT)-enabled monitoring prototype built around an ESP32 microcontroller. The proposed system combines sensor acquisition, local display and alerts, and wireless transfer of selected readings to a monitoring interface. The educational prototype focuses on embedded monitoring, data communication, and safety-oriented control logic. For bench testing, any EMS-related output is represented by a resistive test load rather than stimulation electrodes connected to a person. Sensor readings, alarm thresholds, and communication performance must be validated experimentally and documented before results are claimed.

Keywords: ESP32, Internet of Things, patient monitoring, embedded systems, sensor interfacing, rehabilitation support, safety interlocks.

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1. INTRODUCTION

The Internet of Things connects embedded devices to networks so that data can be collected, displayed, and reviewed remotely. In healthcare-related engineering projects, this approach can support observation and communication, but it does not replace clinical assessment or professional supervision.

This project investigates an ESP32-based prototype for monitoring selected sensor values and sending them to an IoT dashboard. It is designed as an educational engineering system demonstrating sensor interfacing, microcontroller programming, local alerts, and wireless communication. The project title references EMS, but the safe prototype implementation concentrates on monitoring and bench-tested control signals.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1 Problem statement

People with paralysis or reduced mobility may require long-term rehabilitation and monitoring. Conventional monitoring can require repeated in-person checks, while basic standalone devices may not provide convenient remote visibility. A low-cost connected prototype can demonstrate how sensor readings and system status could be made available to an authorized observer.

2.2 Objectives

• Interface selected sensors with an ESP32 development board.

• Display sensor values and system status locally where supported by the hardware.

• Transmit appropriate monitoring data over Wi-Fi to an IoT dashboard or cloud endpoint.

• Provide configurable warning indications using an LED or buzzer for bench demonstrations.

• Develop control logic with timeout, fault-state handling, and clear status reporting.

• Evaluate sensor readings, connectivity, and alert behavior using a documented test plan.

3. PROPOSED SYSTEM

The proposed system consists of a sensor layer, an ESP32 processing and communication layer, a local indication layer, and an IoT dashboard. Sensor values are read at defined intervals, checked for invalid or out-of-range readings, displayed locally, and transmitted when network connectivity is available. If a reading crosses a configured demonstration threshold, the system can activate a local alert. Thresholds used in a student prototype are not medical decision limits.

For the current safe prototype, the EMS output is represented by a resistive test load. No pulse generator or stimulation electrodes should be connected to a person. The ESP32 may demonstrate timing, enable/disable logic, and fault handling without delivering stimulation to a human or animal.

4. HARDWARE AND SOFTWARE REQUIREMENTS

Component / Tool	Purpose
ESP32 development board	Main controller; reads sensors and manages Wi-Fi communication.
MAX30100 sensor (if used)	Demonstration heart-rate and SpO■ sensing; readings require validation.
DS18B20 temperature sensor	Temperature measurement for prototype monitoring.
EMG sensor (if used)	Demonstration of muscle-signal acquisition; requires suitable signal conditioning.
LCD display (if used)	Local display of readings and system status.

Component / Tool	Purpose
LED and buzzer	Local status and warning indication.
Resistive test load	Safe bench representation of an output load; not a patient connection.
Breadboard, wires, regulated supply	Prototype assembly and powering.
Arduino IDE	Firmware development and upload.
IoT dashboard / cloud service	Remote visualization of transmitted test data.

5. SYSTEM DESIGN AND ARCHITECTURE

5.1 Functional block flow

Sensor inputs	ESP32 controller	Outputs / communication
Temperature sensor Optional EMG / MAX30100	Read values Validate data Handle faults	LCD / LED / buzzer Wi-Fi / IoT dashboard
	Bench control logic	Resistive test load only

Data flow: Sensors → ESP32 → local display and alerts → Wi-Fi → IoT dashboard. The dashboard is for observation and is not a substitute for clinical monitoring equipment.

5.2 Main functions

• Acquire sensor readings at a defined sampling interval.

• Check for disconnected sensors, invalid values, and communication errors.

• Display readings and connection state locally.

• Send selected data to the configured IoT service using its supported interface.

• Enter a safe fault state when the system detects a defined error.

6. METHODOLOGY AND IMPLEMENTATION

Step 1 — Hardware preparation

Identify the board and sensor pinouts from their official datasheets. Check supply-voltage requirements before wiring. Use a current-limited, appropriate low-voltage supply for bench work.

Step 2 — Sensor interfacing

Connect one sensor at a time and run a basic example sketch. Compare the output with a reference instrument where available. Record the test conditions and any calibration performed.

Step 3 — Firmware development

Implement sensor initialization, periodic readings, input validation, serial logging, and local status indications. Handle missing or implausible values explicitly.

Step 4 — IoT communication

Configure the selected dashboard credentials locally and transmit test values over Wi-Fi. Avoid committing passwords, access tokens, personal information, or private network credentials to a public repository.

Step 5 — Fault handling

Define demonstrable fault states such as sensor disconnected, Wi-Fi unavailable, or timeout exceeded. Confirm that faults are reported and the bench output remains disabled when required.

Step 6 — Integration

Combine verified modules and repeat tests after integration. Save the firmware, wiring diagram, test records, and photos in the project repository.

7. TESTING AND VALIDATION PLAN

The following table is a test plan, not a claim that the tests have already passed. Enter measured results only after carrying out each test.

Test	Procedure	Result to record
Power-up	Power the ESP32 and inspect startup logs.	Startup status / faults
Sensor response	Compare sensor output with a reference where possible.	Measured value and reference
Sensor disconnect	Disconnect a sensor during bench testing.	Fault detection and response
Wi-Fi loss	Disable the access point temporarily.	Offline behavior and recovery
Dashboard update	Send test values and inspect the dashboard.	Update success and delay
Local alert	Apply a simulated threshold-crossing input.	LED/buzzer response
Safe output state	Trigger timeout or fault logic with the resistive test load connected.	Output state and shutdown behavior

Results section: Complete this section with actual measurements, test dates, hardware versions, and observed limitations. Do not report simulated or expected results as experimental findings.

8. SAFETY, LIMITATIONS AND ETHICAL CONSIDERATIONS

This project is an educational prototype, not a certified medical device. Sensor modules commonly used in hobby projects may not be clinically accurate. The prototype does not establish that EMS prevents or treats muscle atrophy. Electrical stimulation can cause harm if improperly designed or used.

• Do not connect a prototype stimulation output or experimental electrodes to a person or animal.

• Use only a resistive test load for output demonstrations and keep the output disabled by default.

• Use appropriate current limiting, isolation, enclosure, and fault protection in any future engineering evaluation.

• Do not make treatment recommendations from prototype sensor readings.

• Protect credentials and any sensitive data; use fictional test data for public demonstrations.

• Any future human-use investigation requires qualified clinical and electrical-safety oversight, ethics approval where applicable, and compliance with relevant medical-device requirements.

9. EXPECTED OUTCOMES

• A working ESP32 sensor-monitoring demonstration on a bench.

• Local display and alert behavior for selected test conditions.

• Successful or documented-failure IoT data transfer to a dashboard.

• A repository containing firmware, diagrams, documentation, and test records.

• A test report identifying limitations and future improvements.

10. FUTURE SCOPE

• Improve the enclosure, wiring, power management, and maintainability.

• Add secure communication, authenticated access, and better error logging.

• Develop a dashboard with timestamps, graphs, and exportable test data.

• Perform calibration and repeatability studies using suitable reference equipment.

• Undertake formal risk analysis and standards-based verification before considering any medical application.

11. CONCLUSION

The project presents an educational approach to combining ESP32-based sensor monitoring, local alerts, and IoT communication in a rehabilitation-support concept. Its key engineering contribution is the integration of embedded sensing, data handling, connectivity, and fault-aware bench control. Further calibration, documented testing, risk analysis, and expert review are required before making claims about clinical use or effectiveness.

REFERENCES

• Espressif Systems, ESP32 Series Datasheet and Technical Reference documentation.

• Arduino, Arduino IDE documentation and language reference.

• Maxim Integrated / Analog Devices, MAX30100 Pulse Oximeter and Heart-Rate Sensor IC datasheet (if the module is used).

• Analog Devices, DS18B20 Programmable Resolution 1-Wire Digital Thermometer datasheet (if the sensor is used).

• The official datasheets and documentation for each exact sensor module, display, power supply, and IoT service selected for the prototype.

Prepared as a student project report draft. Update component details, wiring, screenshots, measured results, guide information, and institutional formatting to reflect the actual project.